

Hyper-local Circular Economy Platform - AI & Ollama Summary

Project Context

Tech stack: React, Spring Boot, PostgreSQL, FastAPI AI Service, planned Redis integration. Current AI/ML features include Category Prediction (RandomForest) and Price Prediction (XGBoost).

Hardware

Laptop specifications: Intel i5-12450H, 16 GB DDR4 RAM, RTX 3050 4 GB GPU. Capable of running Qwen 3 1.7B, Qwen 3 4B and similar local models through Ollama.

Ollama Setup Findings

Installed Ollama successfully and tested Qwen 3 1.7B and Qwen 3 4B. Qwen 3 models support reasoning ('thinking') mode, which significantly increases response time.

Performance Investigation

Terminal testing with reasoning disabled achieved approximately 1–2 second responses. API-based testing consistently took around 5 seconds. Measured results showed model evaluation around 2.5 seconds and overall request time around 5 seconds.

Category Suggestion

Use case: User enters an item name and receives an automatic category suggestion. Examples: - Canon EOS 1500D DSLR Camera -> ELECTRONICS - Nike Air Max Running Shoes -> FASHION - Bosch Electric Drill Machine -> TOOLS Recommendation: Benchmark Qwen 1.7B against the existing RandomForest model on real item names and choose the more accurate approach.

Description Generator

Generate marketplace-ready item descriptions using title, category, and condition. This is one of the highest-value AI features because users expect AI-generated content to take a few seconds.

Auto Tagging

Generate search-friendly tags such as: camera, dslr, canon, photography, lens Tags improve search quality and can be generated in the background when listings are created.

Search Assistant

Future enhancement that converts natural-language searches into structured categories and keywords for improved listing discovery.

Recommended AI Architecture

Spring Boot -> FastAPI AI Service -> Ollama -> Qwen Model Preferred endpoints: - POST /ai/category - POST /ai/listing-enhancement - POST /ai/search-assistant The listing-enhancement endpoint should generate both descriptions and tags in a single LLM call.

Model Warm-Up Strategy

When FastAPI starts, send a warm-up request to Ollama so the model remains loaded and ready for user requests.

Key Conclusion

The project already has a strong software-engineering foundation with Spring Boot, React, JWT security, RBAC, trust score system, marketplace workflows, and ML integration. Adding Description Generation, Auto Tagging, and Search Assistant capabilities through Ollama will significantly strengthen the project for software engineering and AI-enabled software engineering interviews.